Time Horizon Weather Prediction Machine Learning Model

A Comparative Analysis of ML Techniques for Multi-Step Forecasting

Author: Sehen Elzem Altun

Faculty of Computer Science and Engineering, University of Bologna

The Evolution of Weather Forecasting

Traditional NWP

Numerical Weather Prediction (NWP) relies on fluid dynamics and complex physical principles. It requires **supercomputers** and extreme expertise in atmospheric physics.

The ML Paradigm

Machine Learning models learn patterns from **historical data** without solving physical equations. This offers a computationally efficient alternative for local forecasting.

Core Research Focus

Multi-Horizon

Examining performance degradation across 3-day, 5-day, and 7-day future time frames.

Multi-Output

Simultaneously predicting both Daily Maximum and Minimum air temperatures.

Comparison

Evaluating Linear Regression, Random Forest, SVR, and Multi-Layer Perceptrons.

Methodological Framework: CRISP-DM

- **Data Understanding:** Analysis of meteorological patterns in Cesena, Italy (2015-2025).
- **Data Preparation:** Aggregation of 6-hour measurements into daily max/min features.
- **Modeling:** Implementation of four regression architectures with hyperparameter tuning.
- **Evaluation:** Testing on unseen 2023-2025 data using RMSE, MAE, and R^2 metrics.

Study Area: Cesena, Italy

Coordinates: 44.18° N, 12.21° E

Period: 2015 – 2025

Source: Open-Meteo Platform

The dataset includes 16,072 observations of variables like humidity, pressure, cloud cover, and wind speed, aggregated into a 4,018-sample daily dataset.

Feature Engineering & Reduction

Multicollinearity Removal

Highly collinear features (>0.9 correlation) were removed to prevent instability:

- Temp vs Apparent Temp (0.991)
- Surface Pressure vs MSL Pressure (1.000)
- Soil Temp vs Apparent Temp (0.980)

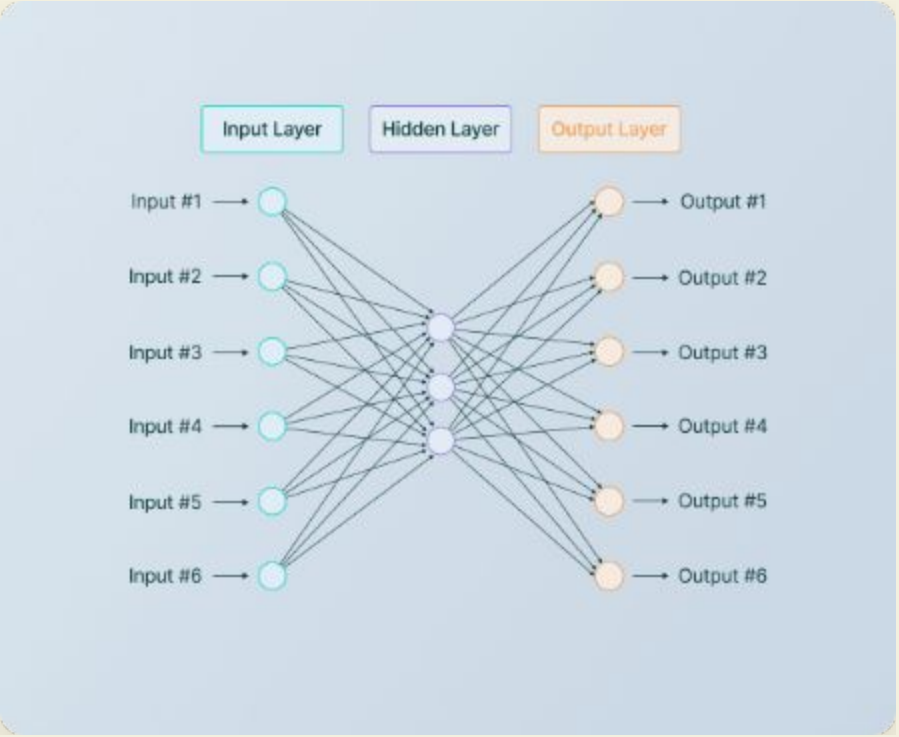
Temporal Decomposition

DateTime stamps were split into discrete numerical features to capture seasonality:

Hour, Month, Year, Day of Week, Day of Month.

Standardization: All features centered at zero mean with unit variance for stable convergence.

Evaluated Architectures



- **Linear Regression:** Statistical baseline model.
- **Random Forest:** Optimized with 200 trees and max depth of 20 via Grid Search.
- **SVR:** RBF kernel with C=100 and Gamma=0.1.
- **MLP (Neural Network):** (100, 50) hidden layer config, ReLU activation, L2 regularization (0.001).

Criteria for Success

Root Mean Square Error

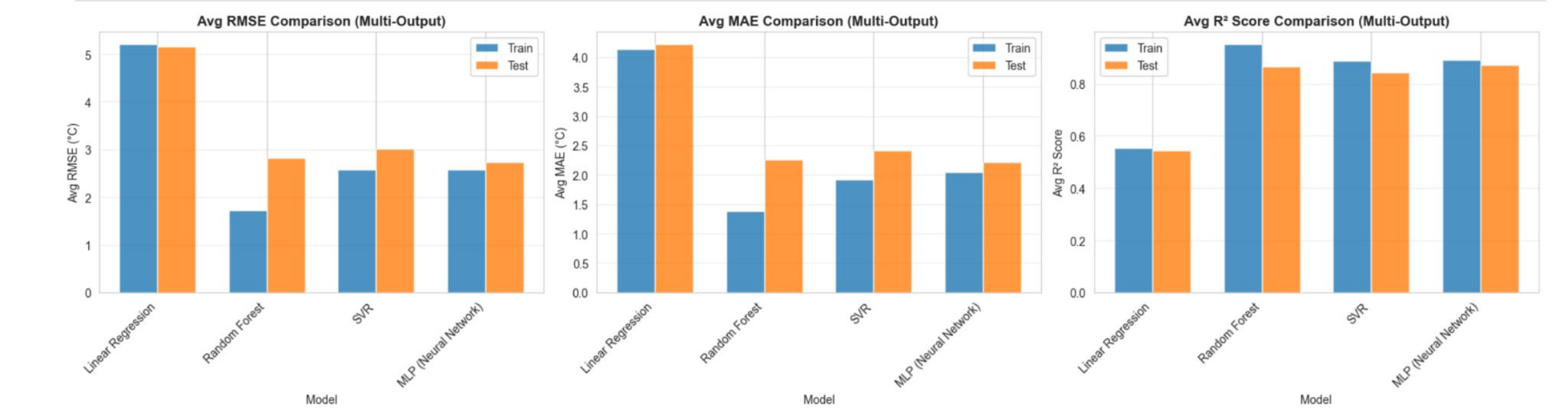
$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Coefficient of Determination

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Metrics are averaged across both Maximum and Minimum temperature outputs.

Same-Day Prediction Performance

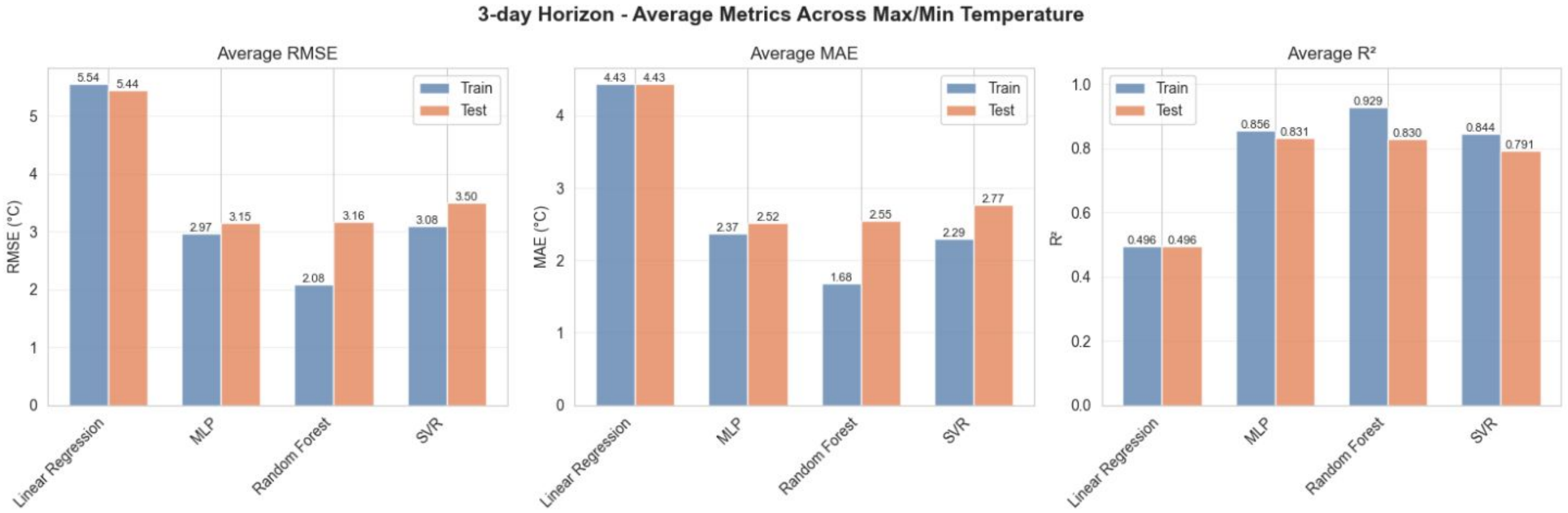


Random forest model has the lowest training RMSE, however, test results shows that there is a slight overfitting

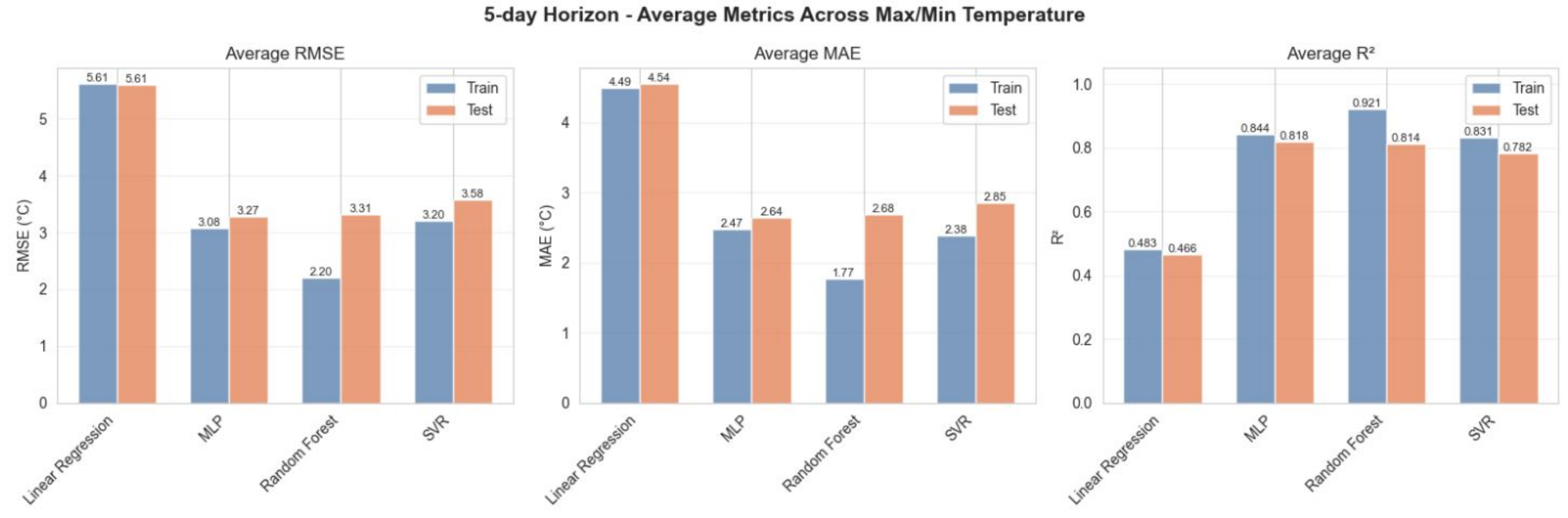
Same-Day Prediction Test Data Performance

MLP achieved a 47% reduction in error compared to the linear baseline.

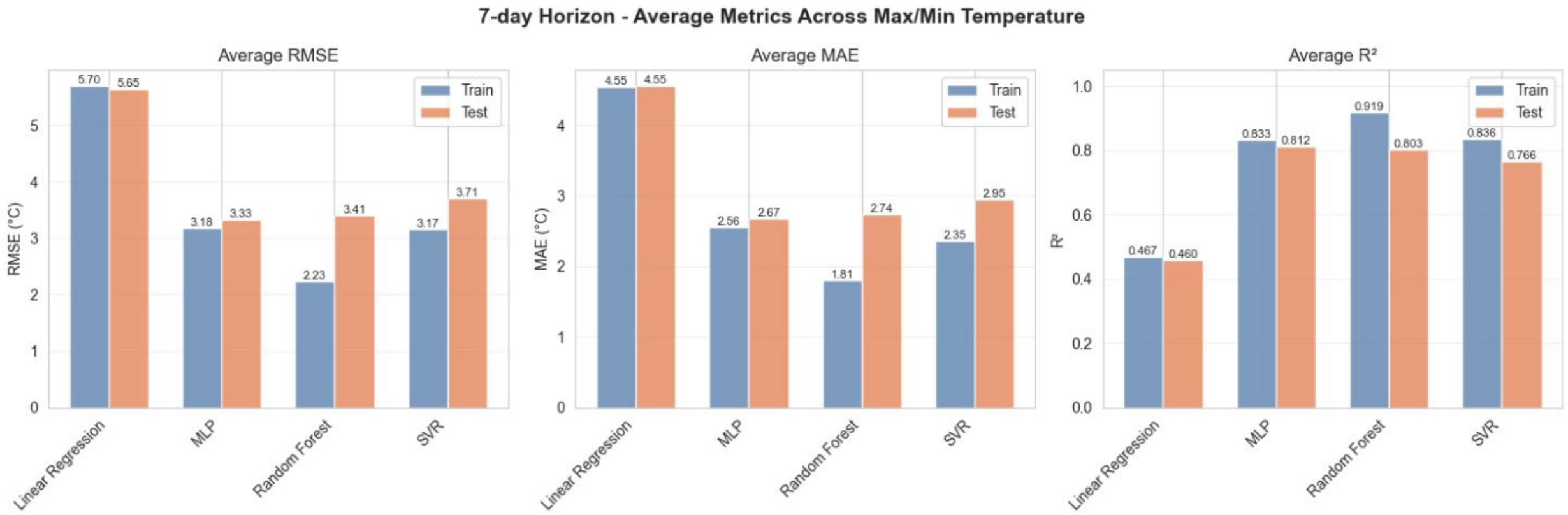
Multi-Step Horizon Forecasting



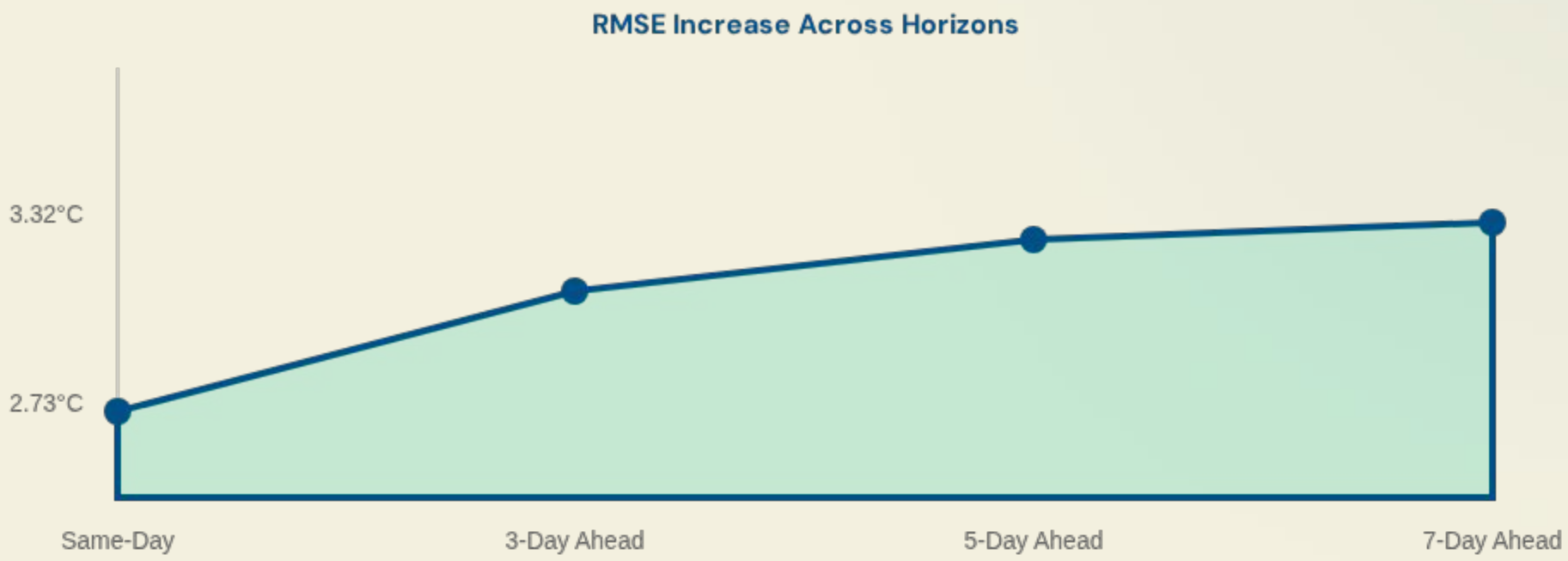
Multi-Step Horizon Forecasting



Multi-Step Horizon Forecasting



Multi-Step Horizon Forecasting (MLP)



Error increase diminishes at longer horizons as models converge toward climatological means.

Key Conclusions

87%
VARIANCE EXPLAINED (R^2)

3.3
MAX RMSE FOR 5-DAY

- **Non-Linear Superiority:** Neural networks are exceptionally well-suited for correlated multi-output tasks.
- **Reliability Window:** Accurate forecasting is highly viable within a 3-5 day window (RMSE < 3.3°C).
- **Efficiency:** Batch learning with structured features avoids the computational complexity of recurrent models.

Thank You

Questions & Discussion

Contact: sehenelzem.altun@studio.unibo.it

University of Bologna | 2025

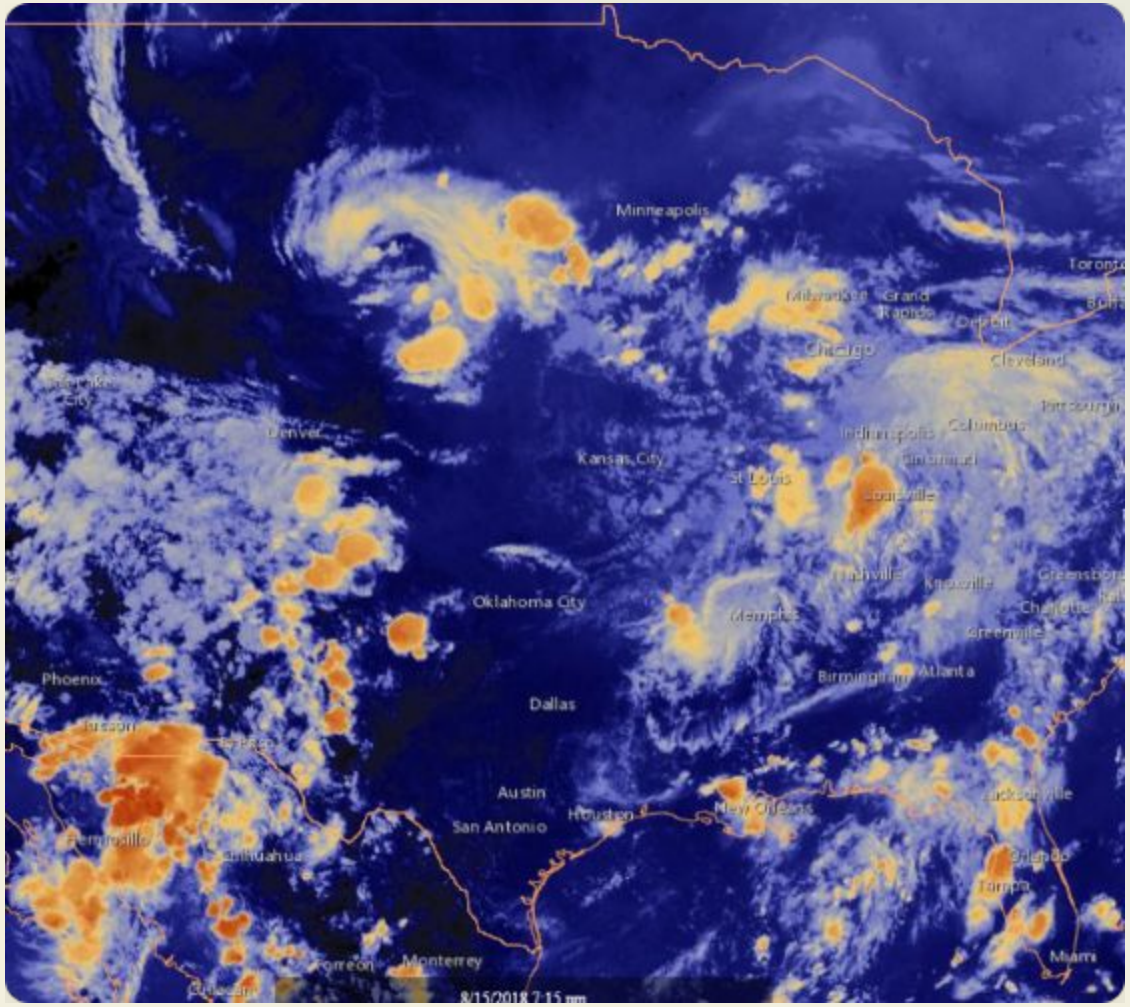


Image Sources

https://static.vecteezy.com/system/resources/previews/060/763/021/non_2x/elegant-abstract-data-analysis-chart-and-graph-transparent-background-professional-free-png.png

Source: www.vecteezy.com

---

https://upload.wikimedia.org/wikipedia/commons/1/1b/Cesena%2C_Emilia-Romagna%2C_Italy.jpg

Source: commons.wikimedia.org

---

https://framerusercontent.com/images/aZW0MrFDu8LFrU2iSYHOIpI8E.png

Source: www.v7labs.com

---

https://www.noaa.gov/sites/default/files/styles/landscape_width_1275/public/legacy/image/2019/Jun/weather_image.jpg

Source: www.noaa.gov